

Lect 15

Recap

1) Expectation & Variance for

example RVs (Bernoulli, Binomial,
geometric)

2) Poisson random variables

- How it approximates # of successful trials for

- a) Independent trials (Binomial PMF)
- b) Otherwise (Matching problem)

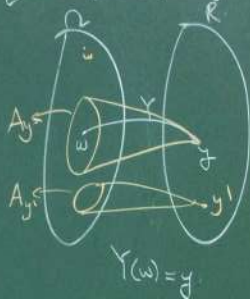
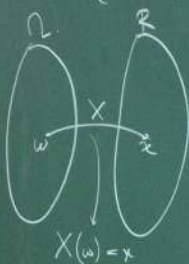
Today

- 1) Joint PMF (multiple RVs) [Marginals from joint PMF]
- 2) Linearity of expectation & examples
- 3) Conditioning.

Joint PMF of two random variables X, Y

$$P_{X,Y}(x,y) = P(\{X=x\} \cap \{Y=y\})$$

$$\{X=x\} = \{\omega \in \Omega \mid X(\omega) = x\}$$



Properties of joint PMF

$$① P_{X,Y}(x,y) \geq 0 \quad \forall x \in \mathcal{X}_0, \forall y \in \mathcal{Y}$$

$$② \sum_{\substack{x \in \mathcal{X}_0 \\ y \in \mathcal{Y}}} P_{X,Y}(x,y) = 1$$

$$③ \left\{ \begin{array}{l} P_X(x) = \sum_{y \in \mathcal{Y}} P_{X,Y}(x,y) \\ P_Y(y) = \sum_{x \in \mathcal{X}_0} P_{X,Y}(x,y) \end{array} \right.$$

marginal PMF

$$③ \Rightarrow ②$$

$$\sum_{x \in \mathcal{X}_0} \left[\sum_{y \in \mathcal{Y}} P_{X,Y}(x,y) \right] = \sum_{x \in \mathcal{X}_0} P_X(x) = 1$$

Property 3 Proof:

$$P_X(x) = P(\{X=x\})$$
$$= P(\{X=x\} \cap \Omega)$$

$$A_y = \{Y=y\} = \{\omega \in \Omega : Y(\omega) = y\}$$

A_y 's partition Ω

$$\Omega = \bigcup_{y \in \mathcal{Y}} A_y$$

$$\Rightarrow P_X(x) = P(\{X=x\} \cap \bigcup_{y \in \mathcal{Y}} A_y)$$
$$= P(\bigcup_{y \in \mathcal{Y}} (\{X=x\} \cap A_y))$$

$$\sum_{y \in \mathcal{Y}} P_{X,Y}(x,y) = \sum_{y \in \mathcal{Y}} P(\{X=x\} \cap \{Y=y\})$$

if $A_1, A_2, \dots, A_n$ are
disjoint,
 $A_1 \cap B, A_2 \cap B, \dots$
are disjoint

Example: Joint PMF of X, Y .

$$X_0 = \{1, 2, 3, 4\} = Y$$

Transportation problem

Exercise: Alternate joint PMF that has same marginal distribution.

$P_{X,Y}(x,y)$

		Y				$P_Y(y)$
		1	2	3	4	
X	1	$1/20$	$2/20$	$3/20$	$1/20$	$3/20$
	2	$1/20$	$3/20$	$2/20$	$1/20$	$7/20$
	3	$1/20$	$1/20$	0	$1/20$	$3/20$
	4					

X	1	2	3	4
$P_X(x)$	$3/20$	$7/20$	$6/20$	$4/20$

Given $P_X(\cdot)$ and $P_Y(\cdot)$ can you find the joint PMF?

Joint PMF of X_1 and X_2

$$P_{X_1, X_2}(x_1, x_2) = \sum_{\substack{\hat{x}_3 \in \mathcal{X}_3 \\ \vdots \\ \hat{x}_n \in \mathcal{X}_n}} P_{X_1, \dots, X_n}(x_1, x_2, \hat{x}_3, \dots, \hat{x}_n)$$

Functions of multiple R.V.s

$Z = g(X, Y)$ You are given $P_{X, Y}(x, y)$

How to find PMF of Z ?

$$P_Z(z) = \sum_{(x, y): g(x, y) = z} P_{X, Y}(x, y)$$

Bernoulli RV(p)

$$\Omega = \{0, 1\}$$

$$X(\omega) = \begin{cases} 1 & \omega \leq p \\ 0 & \text{otherwise} \end{cases}$$

If the sample space Ω is countable

$$E[X] = \sum_{\omega \in \Omega} X(\omega) P(\{\omega\})$$

$$E[X] = \sum_{x \in \mathcal{X}} x P_X(x)$$

$$= \sum_{x \in \mathcal{X}} x$$

$$= \sum_{x \in \mathcal{X}} x \sum_{\substack{\omega \in \Omega \\ X(\omega) = x}} P(\{\omega\})$$

$$= \sum_{x \in \mathcal{X}} \sum_{\substack{\omega \in \Omega \\ X(\omega) = x}} X(\omega) P(\{\omega\})$$



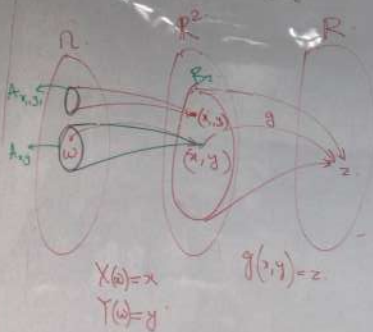
$$= \sum_{\omega \in \Omega} X(\omega) P(\{\omega\})$$

$$= \sum_{x \in \mathcal{X}} x \left(P(\{\omega \in \Omega \mid X(\omega) = x\}) \right)$$

Linearity of Expectation

$$P_z(z) = P(\{\omega: Z(\omega) = z\})$$

$$= P(\{\omega: g(X(\omega), Y(\omega)) = z\})$$



$$B_z = \{(x, y) : g(x, y) = z\}$$

$$\{\omega: g(X(\omega), Y(\omega)) = z\}$$

$$= \bigcup_{\substack{(x,y) \in \mathcal{X} \times \mathcal{Y} \\ g(x,y) = z}} A_{x,y}$$

$$A_{x,y} = \{X=z\} \cap \{Y=y\}$$

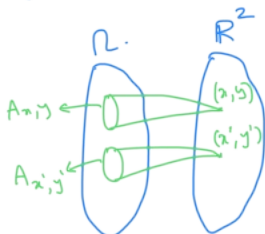
$$\begin{aligned} P_z(z) &= P\left(\bigcup_{\substack{(x,y) \in \mathcal{X} \times \mathcal{Y} \\ g(x,y) = z}} A_{x,y}\right) \\ &= \sum_{\substack{(x,y) \in \mathcal{X} \times \mathcal{Y} \\ g(x,y) = z}} P(A_{x,y}) = \sum_{(x,y): g(x,y) = z} P_{X,Y}(x,y) \end{aligned}$$

Alternate Proof that we tried in class. for

$$Z = g(X, Y), \quad P_Z(z) = \sum_{(x,y): g(x,y)=z} P_{X,Y}(x,y)$$

$$P_Z(z) = P(\{Z=z\}) = P(\{Z=z\} \cap \Omega)$$

$$\Omega = \bigcup_{(x,y) \in \mathcal{X} \times \mathcal{Y}} A_{x,y}$$



$$\Rightarrow P_Z(z) = P(\{Z=z\} \cap \bigcup_{(x,y) \in \mathcal{X} \times \mathcal{Y}} A_{x,y})$$

$$= P\left(\bigcup_{(x,y)} \{Z=z\} \cap A_{x,y}\right)$$

$$= \sum_{x,y} P(\{Z=z\} \cap \{X=x\} \cap \{Y=y\})$$

$$P(\{Z=z\} \cap \{X=x\} \cap \{Y=y\})$$

$$= P(\{\omega: X(\omega)=x, Y(\omega)=y, Z(\omega)=z\})$$

$$= P(\{\omega: X(\omega)=x, Y(\omega)=y, g(X(\omega), Y(\omega))=z\})$$

$$= P(\{\omega: X(\omega)=x, Y(\omega)=y, g(x,y)=z\})$$

$$= \begin{cases} 0 & \text{if } g(x,y) \neq z. \\ P_{X,Y}(x,y) & \text{if } g(x,y)=z \end{cases}$$

$$\Rightarrow P_Z(z) = \sum_{(x,y): g(x,y)=z} P_{X,Y}(x,y).$$

$$E[z] = \sum_{z \in \mathcal{Z}} P_Z(z) z$$

$$= \sum_{z \in \mathcal{Z}} z \left[\sum_{(x,y): g(x,y)=z} P_{X,Y}(x,y) \right]$$

$$= \sum_{z \in \mathcal{Z}} \sum_{\substack{(x,y): \\ g(x,y)=z}} z P_{X,Y}(x,y)$$

$$= \sum_{z \in \mathcal{Z}} \sum_{(x,y): g(x,y)=z} g(x,y) P_{X,Y}(x,y)$$

$$= \sum_{(x,y)} g(x,y) P_{X,Y}(x,y)$$



Linearity of expectation

$$E[a_1 X_1 + a_2 X_2 + \dots + a_n X_n] = a_1 E[X_1] + a_2 E[X_2] + \dots + a_n E[X_n]$$

$P_{X_1, X_2}(x_1, x_2) \rightarrow$ joint PMF

$$E[a_1 X_1 + a_2 X_2] = a_1 E[X_1] + a_2 E[X_2]$$

$$g(x_1, x_2) = a_1 x_1 + a_2 x_2$$

$$E[a_1 X_1 + a_2 X_2] = \sum_{(x_1, x_2)} (a_1 x_1 + a_2 x_2) P_{X_1, X_2}(x_1, x_2)$$

$$= \sum_{x_1, x_2} a_1 x_1 P_{X_1, X_2}(x_1, x_2) + \sum_{(x_1, x_2)} a_2 x_2 P_{X_1, X_2}(x_1, x_2)$$

$$\begin{aligned}
&= \sum_{x_1 \in \mathcal{X}_1} a_1 x_1 \left[\sum_{x_2 \in \mathcal{X}_2} P_{X_1, X_2}(x_1, x_2) \right] + \sum_{x_2 \in \mathcal{X}_2} a_2 x_2 \left[\sum_{x_1 \in \mathcal{X}_1} P_{X_1, X_2}(x_1, x_2) \right] \\
&\quad \downarrow \text{marginal} \qquad \qquad \qquad \downarrow \text{marginal} \\
&= \sum_{x_1 \in \mathcal{X}_1} a_1 x_1 P_{X_1}(x_1) + \sum_{x_2 \in \mathcal{X}_2} a_2 x_2 P_{X_2}(x_2) \\
&= a_1 E[X_1] + a_2 E[X_2]
\end{aligned}$$